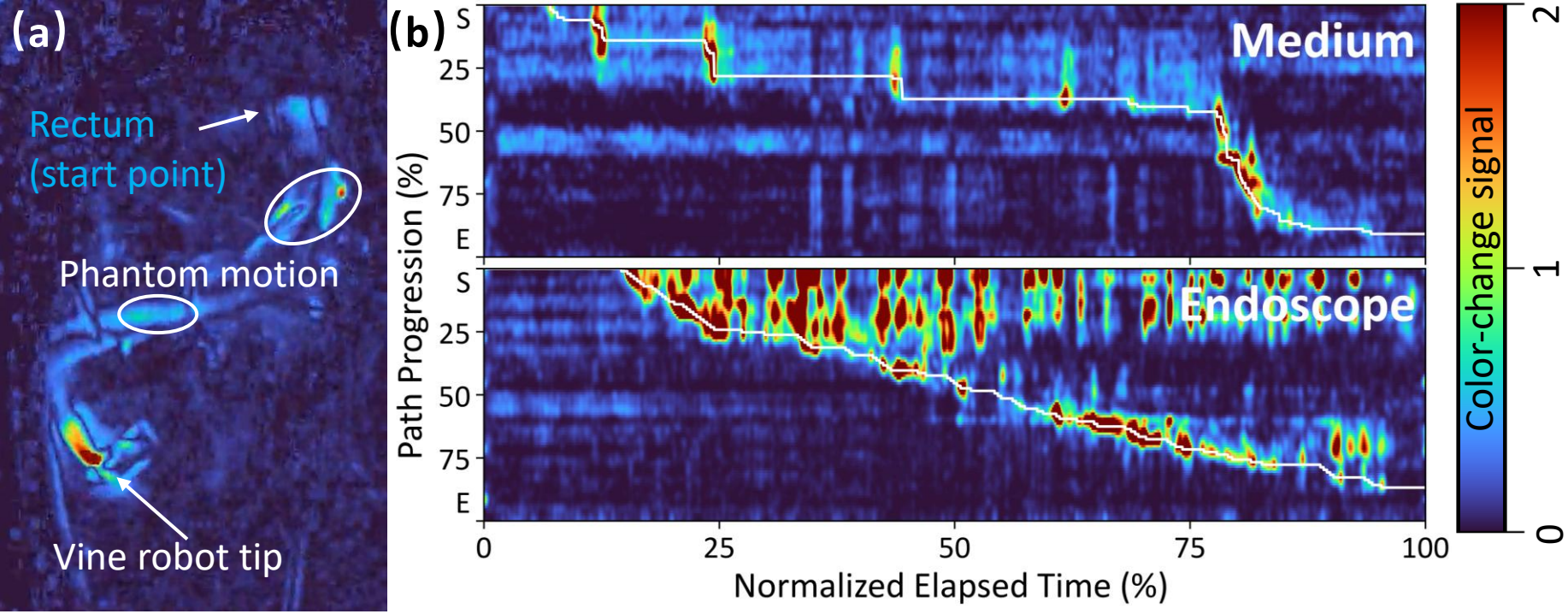

(a)
Rectum
(start point)
Phantom motion
Vine robot tip
(b)
Medium
Endoscope
Path Progression (%)
S
25
50
75
E
0
25
50
75
100
Normalized Elapsed Time (%)
Color-change signal
0
1
2